Band equivalence versus hopping locality: exact rank-two geometry and a smooth continuum benchmark

KSD reduced-Hamiltonian research program

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Abstract

A conventional Bloch calculation cannot distinguish matrix fields related by a momentum-dependent unitary similarity because their bands are identical. The similarity can nevertheless redistribute real-space hopping blocks and produce a nonzero matrix residual when the allowed identification is restricted to one momentum-independent real rotation. We derive the exact adequacy geometry for a rank-two candidate that shifts the energy centroid and positively rescales the traceless level splitting. The spectral loss and every restricted-alignment matrix loss share one diagonal quadratic curvature. Their admissible sets are homothetic ellipses controlled by a splitting scale κ and a normalized correlation μ. We then construct the smooth family $U_a(q) = R(a \sin 2\pi q)$. Although every member has exactly the same bands, its minimum restricted-alignment threshold rises as $\sqrt{\kappa[1 - J_0(2a)^2]}$; under exact spectral acceptance, the joint threshold rises as $\sqrt{2\kappa[1 - J_0(2a)]}$. For a spectrally flat traceless parent, the squared irreducible matrix residual equals the total Fourier weight in nonzero-range hopping blocks generated by the momentum-dependent representation. A pre-version calculation verifies these flat-parent channels under nested momentum refinement and separately exposes unresolved amplitude interpolation. An initial non-flat four-state calculation using the same smooth family preserves roundoff-level band agreement while redistributing existing hopping weight. Its frame-invariant momentum controls converge, while amplitude interpolation remains unresolved. A bounded parent-hopping homotopy shows the mechanism throughout five sampled dispersion scales.

Keywords: reduced Hamiltonian; model adequacy; admissible set; periodic system; spectral error; operator error; convergence; Fourier locality

Experiment identity. The publication-independent experiment is named *Band Equivalence versus Hopping Locality*. Its family identity is `ksdft2effmass.campaign.model-adequacy.periodic1d.band-equivalence-versus-hopping-locality`.

Scope. The threshold continuum and admissible-set formulas are analytic results. A pre-version flat-parent calculation is reported in Section 9 as development analysis, not as retained evidence. Section 10 reports an initial four-state execution whose parent has finite-range hopping, with 'converged' momentum refinement and 'unresolved' amplitude interpolation. The retained Fourier inventories are analyzed without applying a hopping-range truncation. Its parameter bounds, instance-specific acceptance and separation controls, locality ranges, and final numerical tolerances are not frozen. No retained run or accepted instance-specific scientific disposition is asserted here. Non-flat numerical claims are pointwise at declared amplitude nodes; plotted connections are guides to the eye, and continuous-amplitude interpolation is treated as a separate supplementary approximation problem.

1 Introduction

A reduced Hamiltonian can reproduce selected band energies while failing to reproduce a matrix field under a declared identification of retained coordinates. The distinction matters whenever the scientific interpretation depends on matrix elements, finite-range hopping blocks, or locality. Eigenvalues are invariant under a basis rotation, whereas matrix entries and Fourier hopping blocks are not. If the generating momentum-dependent unitary is admitted as the state-space identification, the two matrix fields represent the same abstract operator and the corresponding residual vanishes. The question here is deliberately narrower: can one momentum-independent real rotation identify the fields while retaining their hopping representation? Spectral agreement and this restricted-alignment agreement are related but non-equivalent requirements.

The main question is how strongly these two notions of adequacy can diverge while the Bloch bands remain exactly unchanged. We study a two-band candidate controlled by an energy-centroid shift and a rescaling of the level splitting. The operator comparison additionally allows one global real rotation of the retained basis. A one-parameter family of momentum-dependent target rotations then produces identical bands but a continuously increasing operator-acceptance threshold.

Three outcomes are possible. A *joint match* is established by exhibiting one parameter pair that passes both error limits. *Certified separation* is established only when a verified lower bound places the two admissible parameter regions farther apart than a declared numerical resolution. If neither statement is established, the outcome is *unresolved*. A failed search for a joint match is not a separation proof, and a plot or local optimizer result is not a global certificate.

The central result is an exact reduction of that decision problem. Both losses have the same diagonal quadratic curvature, so the spectral admissible set and every aligned matrix-loss component are homothetic ellipses. Their centers, radii, compatibility condition, and separation distance follow from one splitting scale and one angle-dependent correlation coefficient. For the symmetric rotation family, those quantities yield an explicit monotone threshold curve. The completed flat-parent calculation evaluates the analytic channels without fitting. An initial non-flat implementation evaluates the gap, frame, harmonic, threshold, Fourier, Parseval, and reconstruction channels, while the full admissible-set and post-decision certificate workflow remains prospective.

The computational protocol separates three numerical roles. Construction coordinates determine the loss quadratics, witnesses, certificates, and disposition. Nested construction meshes determine whether those quantities have converged. A final staggered evaluation mesh is used only after the disposition has been fixed and probes behavior between construction nodes. This separation prevents post-decision diagnostics from changing the scientific question that was actually tested.

2 Finite-range periodic realization

2.1 Finite-range parent Hamiltonian

The controlled parent is a one-dimensional periodic model with lattice period a_0 and four states per cell. Let k denote the physical Bloch wavenumber and $G = 2\pi/a_0$ the primitive reciprocal-lattice magnitude. The calculation uses the dimensionless reduced coordinate

$$q = \frac{k}{G} = \frac{ka_0}{2\pi}, \qquad -\frac{1}{2} \le q < \frac{1}{2}. \tag{1}$$

Thus k has units of inverse length, whereas q is expressed in reciprocal-lattice units. The reduced coordinate is used below because lattice translations then enter as $e^{2\pi i q R}$ with integer

cell displacement R. The four states are grouped into a lower two-orbital sector and an upper two-orbital sector. In this block convention, the nonzero hopping matrices are

$$T_0 = \begin{pmatrix} D_{\mathrm{L}} & 0 \\ 0 & D_{\mathrm{U}} \end{pmatrix}, \qquad T_{+1} = \begin{pmatrix} J_{\mathrm{L}} & V \\ 0 & J_{\mathrm{U}} \end{pmatrix}, \qquad T_{-1} = T_{+1}^{\dagger}. \tag{2}$$

In units of the positive energy scale E_G, the 2×2 blocks are

$$D_{\mathrm{L}} = \begin{pmatrix} -1.20 & 0.06 \\ 0.06 & -0.35 \end{pmatrix}, \qquad D_{\mathrm{U}} = \begin{pmatrix} 1.00 & 0.04 \\ 0.04 & 1.75 \end{pmatrix}, \tag{3}$$

$$J_{\mathrm{L}} = \begin{pmatrix} 0.15 & 0 \\ 0 & -0.10 \end{pmatrix}, \qquad J_{\mathrm{U}} = \begin{pmatrix} 0.05 & 0 \\ 0 & -0.04 \end{pmatrix}, \qquad V = \begin{pmatrix} 0.12 & 0 \\ 0 & 0.10 \end{pmatrix}. \tag{4}$$

Thus D_{L} and D_{U} describe intracell mixing within the lower and upper sectors, J_{L} and J_{U} describe nearest-cell hopping within each sector, and V couples the lower sector in one cell to the upper sector in the neighboring cell. Hermitian conjugation supplies the reverse coupling.

With $z = e^{2\pi i q}$, the Bloch Hamiltonian can be read directly in the same convention:

$$K(q) = \begin{pmatrix} D_{\mathrm{L}} + zJ_{\mathrm{L}} + z^{-1}J_{\mathrm{L}}^{\dagger} & zV \\ z^{-1}V^{\dagger} & D_{\mathrm{U}} + zJ_{\mathrm{U}} + z^{-1}J_{\mathrm{U}}^{\dagger} \end{pmatrix} = \sum_{R=-1}^{1} e^{2\pi i q R} T_R. \tag{5}$$

These block values define the parent Hamiltonian rather than one particular momentum sample. All resolution levels evaluate this same parent; refinement changes only the set of sampled momenta. The model is synthetic rather than material-specific. Its two internally mixed orbital sectors are separated in onsite energy but coupled across neighboring cells, producing a momentum-dependent retained subspace with finite hopping support.

2.2 Conventional Bloch-band solution

The analysis begins with the standard Bloch calculation. In operator notation, the cell-periodic Bloch problem is

$$\widehat{K}(k)|\psi_{nk}\rangle = \varepsilon_n(k)|\psi_{nk}\rangle, \qquad k = qG. \tag{6}$$

In the declared four-orbital cell basis $\{|\chi_{\alpha k}\rangle\}_{\alpha=1}^{4}$, Eq. (6) becomes

$$K_{\alpha\beta}(q) = \langle \chi_{\alpha k}|\widehat{K}(k)|\chi_{\beta k}\rangle, \qquad K(q)y_n(q) = \varepsilon_n(q)y_n(q). \tag{7}$$

At each momentum q, the matrix in Eq. (5) is assembled and this Hermitian eigenproblem is solved independently. The four eigenvalues are sorted in ascending order. A conventional band plot consists of the functions $\varepsilon_a(q)$ sampled along the reciprocal mesh.

The present reduction retains the two lowest eigenvectors. The gap between the second and third eigenvalues protects this rank-two subspace from the discarded states. The calculation stops if that gap is smaller than the prescribed lower bound; it does not silently change the retained rank or band identity. At this stage, no relation has been imposed between the eigenvector phases or internal rank-two bases returned at neighboring momenta. Those choices do not affect a conventional band plot because they do not change the eigenvalues.

2.3 Smooth periodic frame

A matrix comparison requires a basis for the retained subspace. Raw eigenvectors from a numerical eigensolver have arbitrary phases and may rotate within the retained rank-two space. The calculation therefore constructs a *discrete parallel-transport gauge* around the reciprocal period. In plain terms, the transport visits momentum points in order:

$$q_0 \longrightarrow q_1 \longrightarrow q_2 \longrightarrow \cdots .$$

At each new point, the raw eigensolver basis is rotated to match the already accepted basis at the preceding point as closely as possible. This removes arbitrary phase changes and basis rotations introduced independently by the eigensolver. Operationally, the procedure is neighbor-to-neighbor SVD/Procrustes alignment. This step fixes the basis convention within the retained subspace; the subspace itself and its spectrum are unchanged.

Let $|y_{nq_i}\rangle$ denote the raw cell-periodic Bloch eigenstates returned for the two retained bands, and let $|f_{nq_i}\rangle$ denote the transported cell-periodic Bloch frame accepted at that point. Collect these kets as the columns of $Y_i \in \mathbb{C}^{4\times 2}$ and $F_i \in \mathbb{C}^{4\times 2}$, respectively. Here $q = k/G$ is the reduced crystal momentum, so a physical one-dimensional wave vector is $k = qG$. Both frames have orthonormal columns. The 2×2 matrix

$$M_i = F_i^\dagger Y_{i+1} \tag{8}$$

is their *neighbor overlap*, with entries $[M_i]_{mn} = \langle f_{mq_i} | y_{nq_{i+1}} \rangle$. Thus the Procrustes problem is the usual Bloch-frame gauge alignment between neighboring crystal momenta, written in finite matrix form.

Write its singular-value decomposition as

$$M_i = L_i \Sigma_i R_i^\dagger, \qquad \Sigma_i = \mathrm{diag}(\sigma_{i,1}, \sigma_{i,2}), \qquad 1 \geq \sigma_{i,1} \geq \sigma_{i,2} \geq 0. \tag{9}$$

The singular values measure the overlap of the two retained subspaces along their two principal directions. A value near one indicates a well-matched direction; a value near zero indicates that one direction at q_{i+1} has almost no overlap with the retained space at q_i.

Starting from $F_0 = Y_0$, the next accepted frame is

$$F_{i+1} = Y_{i+1} Q_i, \qquad Q_i = R_i L_i^\dagger. \tag{10}$$

The unitary matrix Q_i is the rotation that makes the new frame as close as possible to F_i in the Frobenius norm; explicitly,

$$Q_i = \underset{Q^\dagger Q = I}{\mathrm{argmin}} \; \| Y_{i+1} Q - F_i \|_F \,. \tag{11}$$

It changes only the coordinate basis inside the retained subspace; it neither changes the subspace nor refits the Hamiltonian. This is the singular-value form of unitary polar or Procrustes alignment. The real orthogonal least-squares formulation is derived by Schönemann [1]. For the complex unitary form used here, Higham's Eqs. (2.1)–(2.3) construct the unitary polar factor from the singular-value decomposition, and Theorem 2.2, in particular inequality (2.4), gives its Frobenius-norm optimality [2]. The aligned neighbor overlap $F_i^\dagger F_{i+1}$ is Hermitian positive semidefinite.

The protocol introduces a minimum allowed singular value σ_{rank}. The draft value is 0.8; the final value must be recorded in canonical input. Each step must satisfy

$$\sigma_{i,2} > \sigma_{\mathrm{rank}}. \tag{12}$$

If this test fails, the basis at the next point is not sufficiently determined by its overlap with the preceding basis, and the calculation stops.

To state the uniqueness issue explicitly, the polar decomposition of the overlap is $M_i = P_i H_i$, where $P_i = L_i R_i^\dagger$ is unitary and $H_i = R_i \Sigma_i R_i^\dagger$ is Hermitian positive semidefinite. The applied rotation is $Q_i = P_i^\dagger$. When both singular values are positive, P_i and therefore Q_i are unique. If a singular value is zero, rotations acting within the corresponding null directions cannot be inferred from M_i; several equally optimal choices exist. The calculation rejects that case rather than selecting one of those choices. This is the full-rank uniqueness condition in Higham's Theorem 1.1 [2].

Discrete parallel transport can leave a residual mismatch across the periodic boundary. To define that mismatch without an orientation ambiguity, append the duplicate endpoint $Y_N = Y_0$ and apply the same full-rank Procrustes step from F_{N-1} to obtain F_N. Reduce $S = F_0^\dagger F_N$ to its unitary polar factor. If $S = Z\,\mathrm{diag}(e^{i\gamma_1}, e^{i\gamma_2})Z^\dagger$, with principal phases $\gamma_j \in (-\pi, \pi]$, distribute the residual mismatch over the periodic extension by

$$W_i = Z\ \mathrm{diag}\left(e^{-i(i/N)\gamma_1}, e^{-i(i/N)\gamma_2}\right) Z^\dagger, \qquad F_i \leftarrow F_i W_i, \qquad i = 0, \ldots, N. \tag{13}$$

The corrected duplicate endpoint must agree with the corrected start, and all retained sample frames $i = 0, \ldots, N-1$ must satisfy orthonormality and retained-projector reconstruction checks. Proximity to the phase branch cut is recorded as a numerical diagnostic.

The frame remains qualified by the eigensolver and linear-algebra backend that produced it. The protocol does not assert a canonical frame shared by all backends. Projectors, reconstructed operators, and independently verified observables carry the invariant meaning.

2.4 Reference and comparison matrices

The retained reference matrix is

$$H(q_i) = F_i^\dagger K(q_i) F_i. \tag{14}$$

The initial non-flat realization uses the same smooth one-parameter family as the flat benchmark. Define

$$\theta_a(q) = a \sin(2\pi q), \tag{15}$$

$$U_a(q) = \begin{pmatrix} \cos\theta_a(q) & -\sin\theta_a(q) \\ \sin\theta_a(q) & \cos\theta_a(q) \end{pmatrix}, \tag{16}$$

$$A_a(q) = U_a(q)^\dagger H(q) U_a(q), \qquad 0 \le a \le \frac{\pi}{4}. \tag{17}$$

By construction, $A_a(q)$ and $H(q)$ have identical ordered eigenvalues. They are treated as matrix fields in the common retained coordinates defined by $F(q)$. The admissible identification may subsequently apply one global real rotation, but it may not reuse the generating momentum-dependent $U_a(q)$. The resulting loss is therefore a restricted-alignment representation error, not a claim that the two fields cease to be unitarily equivalent as abstract operators. Their Fourier hopping blocks generally differ in the declared coordinates.

The earlier fixed finite-angle field belongs only to the isolated historical regression fixture. Its coefficients are not controls of either active smooth realization.

3 Candidate family and allowed alignment

3.1 Positive trace–traceless affine deformation

Write the retained reference matrix as

$$H(q) = \bar{E}(q) I_2 + D(q), \qquad \bar{E}(q) = \frac{1}{2}\,\mathrm{tr}\,H(q), \qquad \mathrm{tr}\,D(q) = 0. \tag{18}$$

The candidate family is

$$C_{s,\lambda}(q) = \bar{E}(q) I_2 + s E_G I_2 + \lambda D(q), \qquad \lambda > 0. \tag{19}$$

This is the *positive trace–traceless affine deformation family*. Equivalently, it is an energy-centroid shift followed by a rescaling of the level splitting. The parameter s translates both

retained energy levels by sE_G, while λ rescales their displacement from the mean. Positivity of λ preserves the ordering of the two levels.

If $h_1(q) \le h_2(q)$ are the eigenvalues of $H(q)$, the candidate eigenvalues are known without a further eigensolve:

$$c_a(q; s, \lambda) = \bar{E}(q) + sE_G + \lambda[h_a(q) - \bar{E}(q)]. \tag{20}$$

The parameter vector $p = (s, \lambda)^T$ is restricted to a closed rectangle

$$X = [s_{\min}, s_{\max}] \times [\lambda_{\min}, \lambda_{\max}], \qquad \lambda_{\min} > 0. \tag{21}$$

The four bounds affect the scientific outcome and must be fixed before confirmatory execution.

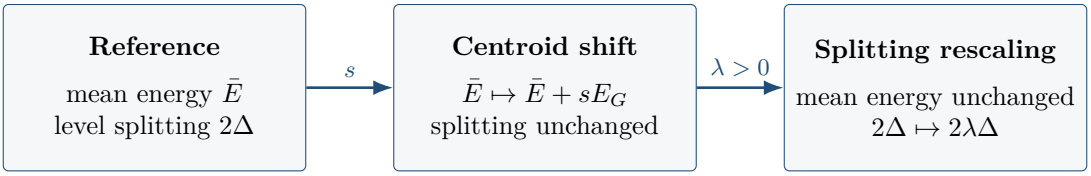


Figure 1: Action of the two candidate parameters. A subsequent basis rotation changes the matrix used for the operator comparison but leaves these energy levels unchanged. The diagram is schematic and contains no calculated data.

3.2 Continuous global alignment family

The operator comparison allows one global real rotation

$$R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}, \qquad C_{s,\lambda,\phi}(q) = R(\phi)^\dagger C_{s,\lambda}(q) R(\phi). \tag{22}$$

The same angle is used at every momentum. Momentum-dependent and unrestricted complex unitary transformations are outside the claim. Since conjugation by $R(\phi + \pi) = -R(\phi)$ is unchanged, the continuous fundamental interval is

$$-\frac{\pi}{2} \le \phi < \frac{\pi}{2}. \tag{23}$$

The optimum over this interval is calculated analytically from the second-harmonic correlation derived below. No finite angle inventory defines the result.

4 Exact quadratic theory

4.1 Loss definitions

Let q_i be a construction coordinate and $w_i > 0$ its raw objective weight. Define

$$\alpha_i = \frac{w_i}{\sum_j w_j}.$$

The canonical input stores the raw weights. A coordinate can be omitted only through a prospective exclusion rule; a zero weight is not an admissible substitute for exclusion.

For retained rank two, the spectral loss and restricted-alignment matrix loss (the operator channel) are

$$L^2_{\mathrm{spec}}(s, \lambda) = \sum_i \frac{\alpha_i}{2} \sum_{n=1}^{2} \left| \frac{c_n(q_i; s, \lambda) - h_n(q_i)}{E_G} \right|^2, \tag{24}$$

$$L^2_{\mathrm{op},\phi}(s, \lambda) = \sum_i \frac{\alpha_i}{2} \left\| \frac{C_{s,\lambda,\phi}(q_i) - A_a(q_i)}{E_G} \right\|_F^2. \tag{25}$$

To connect the matrix loss to the Bloch states, define the retained-space residual operator $\widehat{\Delta H}_{s,\lambda,\phi,a}(k_i) = \widehat{C}_{s,\lambda,\phi}(k_i) - \widehat{A}_a(k_i)$. In the transported Bloch frame,

$$[C_{s,\lambda,\phi}(q_i) - A_a(q_i)]_{mn} = \langle f_{mq_i} | \widehat{\Delta H}_{s,\lambda,\phi,a}(k_i) | f_{nq_i} \rangle. \tag{26}$$

Hence the operator channel is the Brillouin-zone weighted root-mean-square of all retained Bloch-Hamiltonian matrix-element errors, not merely an eigenvalue error. Here and throughout,

$$\|M\|_F^2 = \mathrm{tr}(M^\dagger M) = \sum_{m,n} |M_{mn}|^2 \tag{27}$$

is the squared Frobenius norm: the sum of the squared magnitudes of every matrix entry. The factor $1/2$ gives a per-retained-state normalization, and division by E_G^2 makes both losses dimensionless.

Define the positive splitting scale

$$\kappa = \sum_i \frac{\alpha_i}{2E_G^2} \|D(q_i)\|_F^2 \,. \tag{28}$$

The nontrivial case has $\kappa > 0$, meaning that the retained Hamiltonian is not a scalar multiple of the identity at every construction coordinate.

For one declared amplitude a, suppress the amplitude index on the comparison target. For an allowed global angle ϕ, write

$$D_{i,\phi} = R(\phi)^\dagger D(q_i) R(\phi), \tag{29}$$

$$B_i = U_a(q_i)^\dagger D(q_i) U_a(q_i), \tag{30}$$

$$b_\phi = \sum_i \frac{\alpha_i}{2E_G^2} \mathrm{Re}\,\mathrm{tr}(D_{i,\phi}^\dagger B_i), \tag{31}$$

$$\mu_\phi = \frac{b_\phi}{\kappa}. \tag{32}$$

The scalar μ_ϕ is the normalized correlation between the globally aligned traceless candidate and the momentum-dependent comparison target.

For a traceless Hermitian 2×2 matrix, write

$$D_i = x_i\sigma_x + y_i\sigma_y + z_i\sigma_z, \qquad B_i = X_i\sigma_x + Y_i\sigma_y + Z_i\sigma_z.$$

Conjugation by a real rotation leaves the σ_y component fixed and rotates the (σ_x, σ_z) components through angle 2ϕ. Consequently,

$$b_\phi = b_0 + c\cos(2\phi) + d\sin(2\phi), \tag{33}$$

where

$$b_0 = \sum_i \frac{\alpha_i}{E_G^2} y_i Y_i, \tag{34}$$

$$c = \sum_i \frac{\alpha_i}{E_G^2} (x_i X_i + z_i Z_i), \tag{35}$$

$$d = \sum_i \frac{\alpha_i}{E_G^2} (x_i Z_i - z_i X_i). \tag{36}$$

The exact continuous maximum is

$$b_{\max} = b_0 + \sqrt{c^2 + d^2}, \qquad \mu_{\max} = \frac{b_{\max}}{\kappa}. \tag{37}$$

When $c^2 + d^2$ exceeds the declared degeneracy cutoff, a canonical maximizing angle is

$$\phi_* = \frac{1}{2}\,\mathrm{atan2}(d, c) \quad \text{reduced to} \quad [-\pi/2, \pi/2). \tag{38}$$

If the harmonic amplitude is numerically zero, every angle is equally optimal; the maximum value remains defined but no unique angle is reported.

4.2 Common-curvature theorem

Theorem 1 (Exact loss geometry)**.** *Under Eqs.* (18)–(32)*, the losses are*

$$L^2_{\mathrm{spec}}(s,\lambda) = s^2 + \kappa(\lambda-1)^2, \tag{39}$$

$$L^2_{\mathrm{op},\phi}(s,\lambda) = s^2 + \kappa(\lambda-\mu_\phi)^2 + \kappa(1-\mu_\phi^2), \tag{40}$$

and $|\mu_\phi| \le 1$*. Consequently, the spectral loss and every operator loss have the same parameter-space curvature*

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & \kappa \end{pmatrix}. \tag{41}$$

The operator component at angle ϕ *is centered at* $(0,\mu_\phi)$ *and has irreducible minimum* $m_\phi = \kappa(1-\mu_\phi^2)$*. Because the candidate requires* $\lambda > 0$*, minimizing over the continuous global alignment is equivalent to maximizing* b_ϕ*. The continuously aligned loss is therefore the single quadratic*

$$L^2_{\mathrm{op},*}(s,\lambda) = s^2 + \kappa(\lambda-\mu_{\max})^2 + \kappa(1-\mu_{\max}^2). \tag{42}$$

Proof. Let $d_{ia} = h_a(q_i) - \bar{E}(q_i)$. Since $D(q_i)$ is traceless, $d_{i1} + d_{i2} = 0$. Equation (20) gives the normalized spectral residual

$$s + (\lambda-1)d_{ia}/E_G.$$

When its square is summed over the two retained states, the term linear in $s(\lambda-1)$ vanishes because the two d_{ia} sum to zero. The remaining splitting term is

$$\sum_i \frac{\alpha_i}{2E_G^2} \sum_a d_{ia}^2 = \sum_i \frac{\alpha_i}{2E_G^2} \|D(q_i)\|_F^2 = \kappa,$$

which proves Eq. (39).

For the operator channel, Eqs. (17) and (22) give the normalized matrix residual

$$sI_2 + \frac{\lambda D_{i,\phi} - B_i}{E_G}.$$

The cross term containing s vanishes because both $D_{i,\phi}$ and B_i are traceless. Unitary invariance of the Frobenius norm gives $\|D_{i,\phi}\|_F = \|B_i\|_F = \|D(q_i)\|_F$. Expansion therefore yields

$$L^2_{\mathrm{op},\phi} = s^2 + \kappa\lambda^2 - 2b_\phi\lambda + \kappa.$$

Completing the square and using $\mu_\phi = b_\phi/\kappa$ proves Eq. (40). Finally, Cauchy–Schwarz in the weighted Frobenius inner product gives $|b_\phi| \le \kappa$, hence $|\mu_\phi| \le 1$. □

The theorem identifies the theoretical content of the experiment. Momentum sampling is needed only to evaluate κ and the three harmonic coefficients accurately; it does not create a collection of independent parent models. The incompatibility mechanism is $\kappa(1-\mu_{\max}^2)$: it measures the part of the momentum-dependent traceless target that cannot be represented by one continuously optimized global rotation and one positive uniform splitting scale.

5 Exact admissible-set geometry

Let τ_s and τ_o be the spectral and operator error limits. Their values must be fixed from a stated scientific error budget or identified as synthetic benchmark controls. Define

$$\mathcal{A}_s = \{p \in X : L_{\mathrm{spec}}(p) \le \tau_s\}, \tag{43}$$

$$\mathcal{A}_o = \{p \in X : L_{\mathrm{op},*}(p) \le \tau_o\}. \tag{44}$$

The operator ellipse radius in the scaled coordinates is

$$r_* = \sqrt{\tau_o^2 - \kappa(1 - \mu_{\max}^2)}. \tag{45}$$

The operator admissible set is empty when the radicand is negative.

Each point $p = (s, \lambda)^T$ in these sets denotes one concrete Bloch-Hamiltonian model: sE_G is a uniform centroid-energy shift and λ is a positive rescaling of the retained band splitting. Thus $\mathcal{A}_s$ contains candidates whose band energies meet the spectral error budget, while $\mathcal{A}_o$ contains candidates whose retained Bloch-Hamiltonian matrix elements in the declared frame meet the operator error budget after one global rotation. A *joint witness* is the same physical candidate (s, λ) belonging to both sets. A separation certificate proves only that no member of this declared two-parameter candidate family, under the declared global alignment and error limits, satisfies both requirements. It is not a claim that the complete Hamiltonians are physically inequivalent, that no larger model class can work, or that a material phase or topology has been certified. The ellipses live in model-parameter space, not in the Brillouin zone.

Corollary 2 (Compatibility and exact full-ellipse distance)**.** *Assume r_* is real and that the spectral and continuously optimized operator ellipses are contained in the parameter rectangle X. They intersect if and only if*

$$\sqrt{\kappa}\,|1 - \mu_{\max}| \le \tau_s + r_*. \tag{46}$$

Their Euclidean distance in (s, λ) coordinates is

$$\delta_* = \max\left\{0, |1 - \mu_{\max}| - \frac{\tau_s + r_*}{\sqrt{\kappa}}\right\}. \tag{47}$$

Proof. Use the scaled coordinates $x = s$ and $y = \sqrt{\kappa}\lambda$. By Theorem 1, the spectral ellipse becomes a disk of radius τ_s centered at $(0, \sqrt{\kappa})$. The continuously optimized operator ellipse becomes a disk of radius r_* centered at $(0, \sqrt{\kappa}\mu_{\max})$. Two disks intersect exactly when the distance between their centers does not exceed the sum of their radii, which is Eq. (46). Their centers differ only along the splitting coordinate, so transforming the positive disk gap back to (s, λ) coordinates gives Eq. (47). □

Thus the theoretical decision is explicit. A common witness exists exactly when Eq. (46) holds. Strict separation holds for the full ellipses when that inequality fails; a resolution-qualified claim additionally requires

$$\delta_* > \rho_{\text{sep}} \tag{48}$$

with a numerical uncertainty margin.

The containment premise remains scientifically explicit. The formulas above describe full ellipses, not ellipses clipped by X. The calculation verifies all coordinate extrema before using the corollary as the declared certificate. If containment fails, the current evidence policy records *invalid premise* or *unresolved*; it does not silently replace full-ellipse geometry with an unreviewed box-clipped method.

For numerical verification, a general quadratic is still retained in the form

$$\begin{aligned} r(p) &= r_0 + \mathsf{J}p, & L^2(p) &= p^T Q p + 2g^T p + c, \\ Q &= \mathrm{Re}(\mathsf{J}^\dagger W \mathsf{J}), & g &= \mathrm{Re}(\mathsf{J}^\dagger W r_0), & c &= \mathrm{Re}(r_0^\dagger W r_0). \end{aligned} \tag{49}$$

Here r_0 is the stacked complex residual at $p = 0$, J is its affine parameter-response matrix, and W contains the declared momentum, retained-state, and energy-scale weights. For the spectral channel the stacked entries are Bloch-energy residuals; for the operator channel they

are the Bloch-frame matrix elements in Eq. (26). Thus the generic quadratic is an algebraic representation of the stated Bloch residuals, not a separate fitted model. The independently reconstructed coefficients must agree with the analytic structure in Theorem 1: vanishing cross curvature, unit shift curvature, common splitting curvature κ, spectral center $(0, 1)$, continuous operator center $(0, \mu_{\max})$, and agreement of the retained maximizing angle with Eq. (38). These identities provide stronger verification than merely fitting a quadratic to sampled loss values.

6 Operator information omitted by the Bloch spectrum

The conventional calculation in Section 2.2 retains the ordered eigenvalues. Applied to the controlled pair $H(q)$ and $A_a(q)$, it finds exact agreement because

$$A_a(q) = U_a(q)^\dagger H(q) U_a(q)$$

is a unitary conjugation at every momentum. In the candidate family, the same statement appears as

$$L_{\mathrm{spec}}(0, 1) = 0. \tag{50}$$

Thus a band-only calculation accepts the undeformed candidate independently of how strongly $U(q)$ varies with momentum.

The restricted-alignment matrix question contains information discarded by that comparison. After exact continuous maximization of the global angle, Theorem 1 gives the unconstrained best operator residual

$$\inf_{s\in\mathbb{R},\,\lambda\geq 0} L^2_{\mathrm{op},*}(s, \lambda) = \kappa \left[1 - \max(0, \mu_{\max})^2\right]. \tag{51}$$

A strictly positive lower splitting bound can only increase this infimum. When the momentum-dependent comparison field is exactly representable by one global rotation, $\mu_{\max} = 1$ and the residual vanishes at $(s, \lambda) = (0, 1)$. Otherwise, $1 - \mu^2_{\max}$ quantifies the part that a conventional band comparison cannot see and a single global alignment cannot reproduce.

The same distinction appears in real space. If

$$H(q) = \sum_R e^{2\pi i q R} T_R, \qquad U(q) = \sum_m e^{2\pi i q m} U_m,$$

then the hopping blocks of $A(q) = U(q)^\dagger H(q) U(q)$ are

$$\widetilde{T}_Q = \sum_{m,n} U_m^\dagger T_{Q+m-n} U_n. \tag{52}$$

A momentum-independent basis change has only one nonzero Fourier coefficient and conjugates each hopping block without changing its range. A momentum-dependent change mixes displacement indices through Eq. (52); it can redistribute weight to longer-range blocks while leaving every band energy unchanged.

The comparison is therefore not between two ways of computing the same band structure. The conventional calculation answers whether the energies agree. The present theory asks the stronger question of whether the same restricted candidate also reproduces the Hamiltonian in a declared frame and the associated hopping representation. The scalar $\mu_{\max}$ is the exact bridge between those questions for the rank-two family.

6.1 An exact band-only false positive

The theory permits an explicit construction in which the conventional band criterion passes exactly while joint spectral and operator adequacy fails by a positive margin. This is an analytic counterexample, not a calculated outcome for the four-state parent.

Proposition 3 (Exact spectral pass with certified operator incompatibility)**.** *Consider two equally weighted momentum observations with*

$$H_1 = H_2 = \bar{E} I_2 + E_G \sigma_z, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and comparison rotations

$$U_1 = R(0), \qquad U_2 = R(\pi/4).$$

Allow every global real rotation $R(\phi)$ and choose

$$X = [-0.20, 0.20] \times [0.50, 1.10], \qquad \tau_s = 0.05, \qquad \tau_o = 0.72.$$

Then the conventional spectral loss vanishes at $(s, \lambda) = (0, 1)$, but no point in X satisfies both error limits. The Euclidean separation of the closest full admissible ellipses is

$$\delta_* = 1 - \frac{1}{\sqrt{2}} - 0.05 - \sqrt{0.72^2 - \frac{1}{2}} \approx 0.1072466192. \tag{53}$$

Consequently, any declared separation resolution $\rho_{\mathrm{sep}} < \delta_*$ is certified.

Proof. For this construction, $\kappa = 1$. Conjugation of σ_z by a real rotation changes its orientation through twice the rotation angle. Equation (32) therefore gives

$$\mu_\phi = \frac{1}{2}\left[\cos(2\phi) + \cos\left(\frac{\pi}{2} - 2\phi\right)\right] = \frac{1}{\sqrt{2}} \cos\left(2\phi - \frac{\pi}{4}\right).$$

Hence the largest attainable positive correlation is $\mu_{\max} = 1/\sqrt{2}$, reached at $\phi = \pi/8$. The spectral admissible ellipse is

$$s^2 + (\lambda - 1)^2 \le 0.05^2.$$

At the best global angle, the operator ellipse is

$$s^2 + \left(\lambda - \frac{1}{\sqrt{2}}\right)^2 \le 0.72^2 - \frac{1}{2}.$$

Both ellipses lie inside X. Their centers differ only along the splitting coordinate, so Corollary 2 gives Eq. (53).

Angles with smaller positive μ_ϕ cannot reduce the separation: wherever their operator component is nonempty, decreasing μ_ϕ moves its center farther from $\lambda = 1$ and does not compensate for that shift through its radius. Components with negative μ_ϕ are either empty at the stated threshold or lie entirely below $\lambda = 0.50$. Thus no allowed global angle produces a joint witness in X. □

This construction states precisely what "the bands pass but the reduced Hamiltonian fails" means. The failure is not an optimizer failure: it follows from the exact loss geometry and a positive certified distance between admissible sets.

6.2 A smooth continuum of band-equivalent Hamiltonians

For comparison across the complete reciprocal period, use the smooth periodic family already defined in Eq. (17):

$$\mathcal{U}_{\text{smooth}} = \left\{ U_a(q) = R(a \sin 2\pi q) : 0 \le a \le \frac{\pi}{4} \right\}. \tag{54}$$

The parameter a is the maximum basis-rotation amplitude. At $a = 0$, the comparison and reference matrices coincide. Increasing a introduces progressively stronger momentum-to-momentum basis variation while preserving every Bloch eigenvalue exactly. Unlike a piecewise rotation, Eq. (54) is smooth and periodic, so its induced hopping tail reflects a controlled Fourier hierarchy rather than a jump discontinuity.

For the flat traceless rank-two parent in Eq. (59), the Jacobi–Anger expansion gives the reciprocal-period average

$$\mu_{\max}(a) = J_0(2a), \tag{55}$$

where J_0 is the Bessel function of the first kind [3, Sec. 10.12].

Corollary 4 (Smooth monotone threshold continuum)**.** *For the family in Eq.* (54)*, before finite parameter bounds are applied, the infimum operator threshold is*

$$\tau_{o,\min}(a) = \sqrt{\kappa \left[1 - J_0(2a)^2\right]}. \tag{56}$$

For a fixed spectral threshold τ_s, the smallest operator threshold permitting a joint witness is

$$\tau_{o,\text{joint}}(a;\tau_s) = \left[\kappa \left[1 - J_0(2a)^2\right] + \max\left\{0, \sqrt{\kappa}\left[1 - J_0(2a)\right] - \tau_s\right\}^2 \right]^{1/2}. \tag{57}$$

For exact spectral acceptance this reduces to

$$\tau_{o,\text{joint}}(a;0) = \sqrt{2\kappa \left[1 - J_0(2a)\right]}. \tag{58}$$

All three quantities are nondecreasing on $0 \le a \le \pi/4$.

Proof. For $D = E_G \sigma_z$, conjugation through Eq. (54) gives

$$U_a(q)^\dagger D U_a(q) = E_G \left[\cos(2a \sin 2\pi q)\, \sigma_z - \sin(2a \sin 2\pi q)\, \sigma_x\right].$$

The sine term has zero reciprocal-period average, while the cosine average is $J_0(2a)$. A global rotation aligned with σ_z therefore maximizes the operator correlation and proves Eq. (55). Substitution into Theorem 1 gives Eq. (56). Solving the exact disk-intersection condition in Eq. (46) gives Eqs. (57) and (58). On the stated interval, $2a$ remains before the first positive zero of J_0, and $J_0(2a)$ decreases monotonically; the thresholds therefore increase monotonically. □

This family supports a direct side-by-side calculation across a: the conventional branch reports zero band change, while the representation-aware branch reports the operator threshold and the induced Fourier hopping tail.

7 Exact relation to representation-induced hopping range

The threshold has a direct real-space meaning. Consider the spectrally flat traceless parent

$$H(q) = \bar{E} I_2 + D, \qquad D = D^T \in \mathbb{R}^{2\times 2}, \qquad \operatorname{tr} D = 0, \qquad D \neq 0, \tag{59}$$

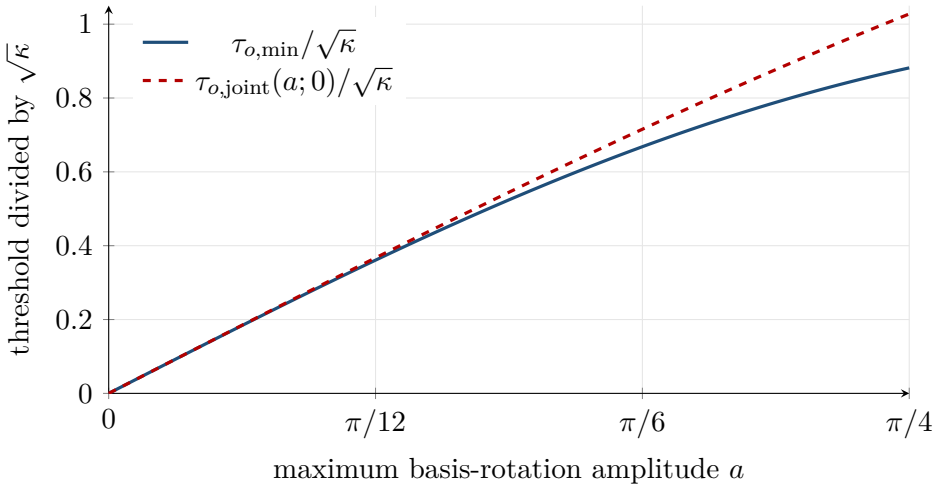


Figure 2: Analytic acceptance thresholds for the smooth continuum in Eq. (54). Every amplitude has exactly the same Bloch bands. The solid curve is the smallest threshold admitting an operator fit before finite parameter bounds are applied. The dashed curve additionally requires exact spectral agreement. The plotted points are evaluations of the displayed Bessel formulas, not empirical data.

where $\bar{E}$ and D are independent of momentum. Its only hopping block is onsite. After a momentum-dependent real rotation, the traceless comparison field is

$$B(q) = U(q)^\dagger D U(q). \tag{60}$$

Define its Fourier blocks by

$$B_R = \int_{-1/2}^{1/2} e^{-2\pi i q R} B(q)\, dq, \qquad B(q) = \sum_{R\in\mathbb{Z}} e^{2\pi i q R} B_R. \tag{61}$$

The nonzero-R blocks are hopping terms created entirely by the momentum dependence of the basis transformation.

Theorem 5 (Operator residual equals induced nonlocal hopping weight)**.** *Let $B(q)$ be given by Eq.* (60)*, with $U(q)$ restricted to the same real two-dimensional rotation family as the allowed global alignment. Set*

$$\kappa = \frac{\|D\|_F^2}{2E_G^2}, \qquad \mu_{\max} = \frac{\|B_0\|_F}{\|D\|_F}. \tag{62}$$

Then

$$\inf_{s\in\mathbb{R},\,\lambda\geq 0,\,\phi} L^2_{\mathrm{op},\phi}(s,\lambda) = \kappa(1-\mu_{\max}^2) \tag{63}$$

$$= \frac{1}{2E_G^2}\sum_{R\neq 0}\|B_R\|_F^2\,. \tag{64}$$

Thus the squared irreducible operator residual is exactly the normalized Fourier weight outside the onsite block. For a finite uniform mesh, the same identity holds with the integral and Fourier series replaced by the corresponding discrete transform.

Proof. Every matrix $U(q)^\dagger D U(q)$ is traceless, has Frobenius norm $\|D\|_F$, and lies in the two-dimensional real rotation orbit of D. Its average B_0 lies in the disk spanned by that orbit. Hence there are a global angle ϕ_0 and a scale $\lambda_0 \geq 0$ such that

$$B_0 = \lambda_0 R(\phi_0)^\dagger D R(\phi_0), \qquad \lambda_0 = \frac{\|B_0\|_F}{\|D\|_F} = \mu_{\max}.$$

The best constant globally aligned field is therefore the orthogonal projection B_0. The trace shift is minimized by $s = 0$, and

$$\inf_{s,\lambda,\phi} L^2_{\mathrm{op},\phi} = \frac{1}{2E_G^2} \int_{-1/2}^{1/2} \|B(q) - B_0\|_F^2 \, dq. \tag{65}$$

Parseval's identity applied entrywise to Eq. (61) turns the right side into the nonzero-mode sum in Eq. (64). Unitary invariance gives $\|B(q)\|_F = \|D\|_F$ pointwise, so a second application of Parseval gives

$$\sum_{R\neq 0} \|B_R\|_F^2 = \|D\|_F^2 - \|B_0\|_F^2 = \|D\|_F^2 \, (1 - \mu_{\max}^2).$$

Division by $2E_G^2$ proves Eq. (63). □

For the smooth continuum in Eq. (54), $\mu_{\max}(a) = J_0(2a)$. The theorem therefore gives

$$\frac{1}{2E_G^2} \sum_{R\neq 0} \|B_R(a)\|_F^2 = \kappa \left[1 - J_0(2a)^2\right] = \tau_{o,\min}(a)^2. \tag{66}$$

The solid curve in Fig. 2 is consequently the square root of the induced non-onsite hopping weight. If exact spectral agreement additionally fixes $\lambda = 1$, the joint threshold contains a second contribution from the mismatch of the onsite block:

$$\tau_{o,\mathrm{joint}}(a;0)^2 = \underbrace{\kappa \left[1 - J_0(2a)^2\right]}_{\text{non-onsite hopping weight}} + \underbrace{\kappa \left[1 - J_0(2a)\right]^2}_{\text{onsite-block mismatch}}. \tag{67}$$

This identity is the substantive distinction from a band-only diagnostic. The band energies remain unchanged, but the momentum-dependent representation necessarily moves a calculable amount of operator weight outside the cell-local block. A global rotation can change the orbital basis within each cell but cannot remove those nonzero-range Fourier blocks.

8 Side-by-side Bloch and representation-aware calculation

The computational study evaluates the same continuum through two branches. Both branches start from one solution of the parent Bloch eigenproblem. The parent is not refitted or rediagonalized as a different physical model at each amplitude.

8.1 Conventional Bloch branch

For every amplitude a and momentum q_i, construct $A_a(q_i)$ from Eq. (54) and independently diagonalize it. Record the ordered-band defect

$$\Delta_{\mathrm{band}}(a) = \max_{i,n} \frac{|\varepsilon_n[A_a(q_i)] - \varepsilon_n[H(q_i)]|}{E_G}. \tag{68}$$

The analytic value is zero for every a; the calculated value measures only numerical eigensolver and ordering error. The branch also records the two overlaid band curves at selected amplitudes. No operator matrix element or hopping block enters its acceptance criterion.

8.2 Representation-aware branch

Using the same matrices $A_a(q_i)$, calculate

1. the best global alignment correlation $\mu_{\max}(a)$;

2. the irreducible operator loss $L_{\mathrm{op,min}}(a)$;

3. the operator threshold required for a joint witness at each declared spectral threshold;

4. the Fourier hopping blocks $\widetilde{T}_R(a)$; and

5. the range-resolved tail

$$P_{>R_c}(a) = \frac{1}{2E_G^2} \sum_{|R|>R_c} \left\| \widetilde{T}_R(a) \right\|_F^2 . \tag{69}$$

For the flat analytic parent, $R_c = 0$ and

$$L_{\mathrm{op,min}}(a)^2 = P_{>0}(a) = \kappa \left[1 - J_0(2a)^2 \right] . \tag{70}$$

The calculation retains the defect in this three-way identity as a verification channel. For the finite-range four-state realization, the parent itself has nonzero-range blocks; the analysis therefore reports the complete range-resolved distribution and its change with a, without assuming that Eq. (70) continues to hold unchanged.

8.3 Continuum sampling and comparison figures

The proposed amplitude interval is $0 \le a \le \pi/4$. A nested amplitude study uses

$$a_j = \frac{j\pi}{4N_a}, \qquad j = 0, \ldots, N_a, \qquad N_a \in \{8, 16, 32, 64\}. \tag{71}$$

Coarse amplitudes reappear on every doubled grid. Convergence is required for $\mu_{\max}$, the operator thresholds, and every reported hopping-tail channel. Comparing only shared amplitude nodes is insufficient because those values agree by construction. A current exploratory midpoint-interpolation analysis finds maximum defects 3.08×10^{-5} and 1.41×10^{-5} for the two threshold curves at $N_a = 64$, above the draft unit-scale tolerance 1.01×10^{-8}. The ladder in Eq. (71) therefore remains a proposal and must not be frozen. This finding does not require a uniform 4096-interval grid if the final protocol instead evaluates the analytic law directly or adopts a reviewed adaptive amplitude rule. The initial non-flat study independently confirms that the same proposed ladder is insufficient: at $N_a = 64$, its largest correlation and hopping-block midpoint defects are 3.49×10^{-5} and 1.39×10^{-5}, respectively. No flat-parent analytic comparator is used for those non-flat defects. The deterministic adaptive action caches direct samples, evaluates interval midpoints, refines only failing intervals, and then requires disjoint quarter-point validation. It remains 'unresolved' when its explicit level or unique-sample cap is reached. Its production cap and thresholds are not frozen.

The composite comparison in Fig. 3 places conventional Bloch bands, operator-acceptance thresholds, amplitude resolution, cumulative tails, and range-resolved hopping weight in one view. The intended contrast is quantitative: the bands remain unchanged, while the representation-aware panels resolve the increasing operator mismatch and representation-induced hopping range.

9 Exploratory flat-parent analysis

Figure 3 summarizes the pre-version calculation for the normalized smooth flat-parent family. It is a development result generated by the current software implementation; it is not a retained run or an evidence package.

The reference and transformed models retain the same Bloch eigenvalues across the Brillouin zone. Their largest calculated band defect is 2.22×10^{-16} [Fig. 3(d)], so the conventional band

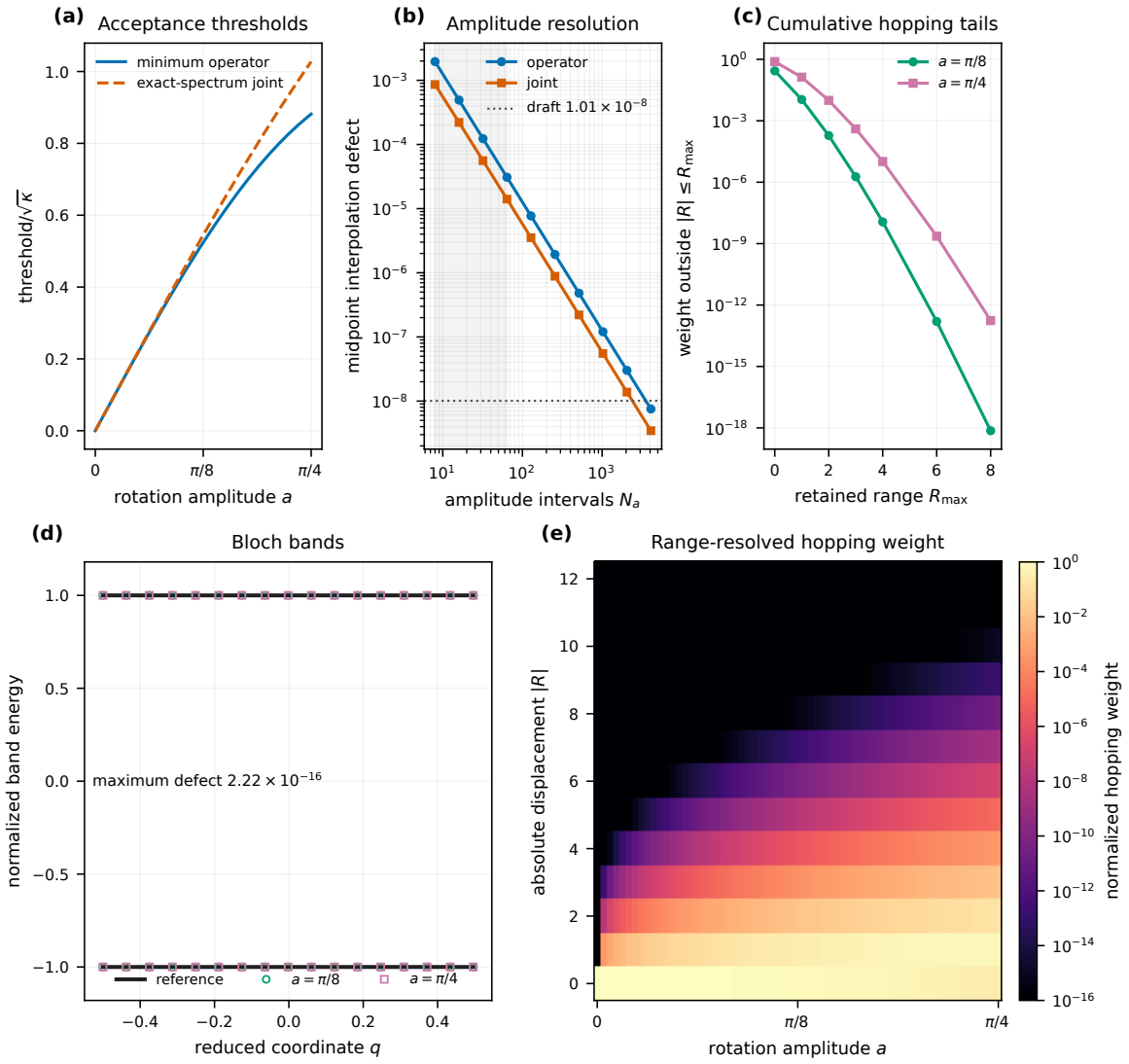


Figure 3: Band equivalence versus hopping locality in the smooth flat-parent continuum. **(a)** Normalized minimum operator and exact-spectrum joint acceptance thresholds as functions of the maximum rotation amplitude a. **(b)** Maximum midpoint defect of piecewise-linear interpolation on nested uniform amplitude grids; the shaded region marks the proposed $N_a = 8$–64 ladder, and the dotted line is the draft 1.01×10^{-8} unit-scale target. **(c)** Cumulative normalized hopping weight outside the retained range $|R| \leq R_{\max}$ for two amplitudes. **(d)** Reference and transformed Bloch bands; the largest computed eigenvalue defect is 2.22×10^{-16}. **(e)** Normalized range-resolved hopping weight over amplitude and absolute displacement. Thresholds in panel (a) use the analytic $J_0(2a)$ comparator; panels (c) and (e) use the $N = 512$ finite transform after nested momentum refinement. The calculation shown is pre-version development analysis and is not a retained evidence run.

structure does not distinguish the two representations. The representation-aware thresholds nevertheless increase continuously with amplitude [Fig. 3(a)]. Equations (56) and (58) give normalized minimum-operator and exact-spectrum joint thresholds of 0.8815978972 and 1.0276174232, respectively, at $a = \pi/4$. Their separation shows that exact band agreement does not remove the operator-level cost of identifying the representations.

The same mismatch appears in real-space locality. At $a = 0$, all normalized hopping weight outside the onsite block is below 1.3×10^{-28} on the $N = 512$ transform. The total non-onsite weight rises to 0.2747231 at $a = \pi/8$ and 0.7772149 at $a = \pi/4$. The cumulative tails in Fig. 3(c) decay rapidly with retained range, but their amplitude dependence is pronounced: the weight outside $|R| \leq 4$ is 1.16×10^{-8} at $a = \pi/8$ and 1.03×10^{-5} at $a = \pi/4$. The range-resolved map in Fig. 3(e) confirms that the support broadens continuously even though the bands remain unchanged. This is the calculated counterpart of the exact flat-parent identity in Eq. (70).

The numerical study separates momentum and amplitude resolution. For the implemented flat-parent channels, refinement from $N = 32$ through $N = 512$ changes every monitored momentum-dependent quantity only at approximately binary64 roundoff. By contrast, the proposed uniform amplitude ladder through $N_a = 64$ is insufficient to reconstruct a continuous threshold curve at the draft 1.01×10^{-8} level [Fig. 3(b)]. Its maximum midpoint interpolation defects are 3.08×10^{-5} and 1.41×10^{-5} for the operator and joint thresholds. The first tested uniform grid to pass the draft target has $N_a = 4096$. This does not imply that every calculation requires 4096 amplitudes: the analytic flat-parent laws can be evaluated directly, while a non-analytic realization should use a reviewed adaptive rule rather than infer between-node convergence from shared nested nodes alone.

This benchmark establishes the analytic mechanism and tests the implemented flat-parent correlation, threshold, band-defect, and Fourier-locality channels, but it does not test the prospective frame, admissible-set, witness, or separation-certificate workflow and does not establish material realism. The parent is deliberately flat, and the reported hopping blocks are finite-mesh Fourier coefficients. A qualified non-flat realization and a frozen retained run remain necessary before these numerical values can serve as final publication evidence.

10 Exploratory non-flat four-state analysis

The initial typed implementation constructs the parent in Eq. (5), selects the lower two-state spectral subspace, closes the parallel-transport frame, and applies Eq. (17). Figure 4 reports the resulting pre-version diagnostics. They are neither retained evidence nor a converged numerical result.

At $N = 512$, the minimum sampled external gap is 1.0566461, the minimum neighboring-frame transport singular value is 0.9999997323, and the largest independently reconstructed result defect is 3.09×10^{-15}. The largest computed band defect over 65 amplitudes is 1.45×10^{-15}. At $a = \pi/4$, the discrete maximum correlation is 0.5079908088, the minimum operator threshold is 0.4004394264, and the exact-spectrum joint threshold is 0.4611609032 [Fig. 4(a)]. The comparison non-onsite hopping weight increases from the reference value 0.0323839143 to 0.1761532501. Its weight outside $|R| \leq 4$ is 8.43×10^{-6}, whereas the reference value is 1.52×10^{-18} [Fig. 4(d)]. The full amplitude-dependent redistribution is shown in Figs. 4(b)–(c). These are finite-mesh observations, not the flat-parent identity in Eq. (70).

The momentum disposition is 'converged'. Between $N = 256$ and $N = 512$, the largest selected outcome-channel change is 1.08×10^{-11} and the largest parent-invariant change is 1.67×10^{-16} [Fig. 4(e)]. Neighboring-frame overlap remains a qualification lower bound rather than a convergence observable; shared parent eigenvalues and retained spectral projectors provide the invariant cross-mesh checks. At $N_a = 64$, the largest correlation midpoint defect is

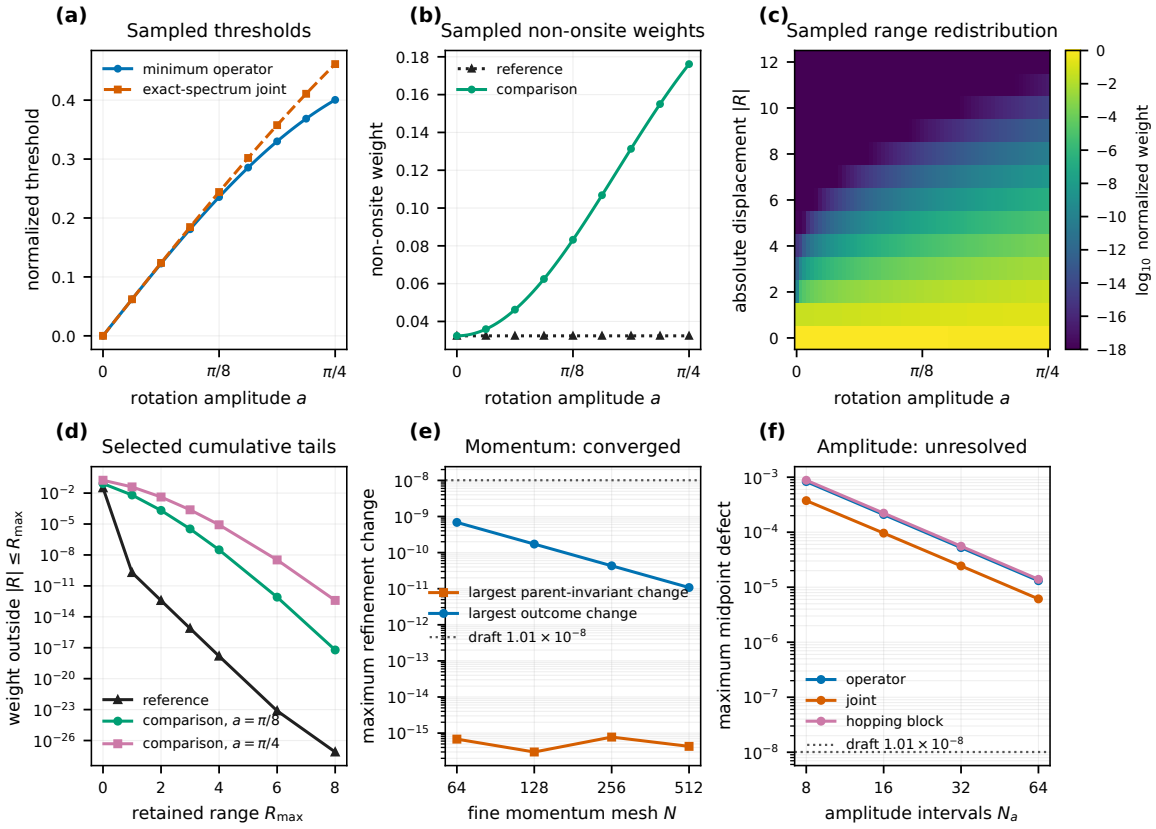


Figure 4: Exploratory smooth non-flat four-state realization. **(a)** Restricted-alignment minimum-operator and exact-spectrum joint thresholds over a 65-node sampling of the smooth amplitude family. Markers show every eighth direct sample and connecting lines are guides to the eye. **(b)** Reference and comparison non-onsite weights over the same samples. **(c)** Range-resolved comparison hopping redistribution over amplitude and absolute displacement. **(d)** Selected cumulative finite-mesh hopping tails. **(e)** Adjacent momentum-refinement changes: both parent-invariant and selected outcome channels lie below the draft target. **(f)** Midpoint amplitude-interpolation defects, all above the draft target on the tested $N_a \leq 64$ ladder. No $J_0(2a)$ comparator or flat-parent residual–tail identity is used. Momentum refinement is 'converged', amplitude interpolation is 'unresolved', and the calculation is not a retained evidence run.

3.49×10^{-5} and the largest hopping-block-power midpoint defect is 1.39×10^{-5} [Fig. 4(f)]. The failing amplitude control is not overridden by momentum convergence, roundoff-level band agreement, or apparently stable selected thresholds.

10.1 Bounded parent-family robustness study

To distinguish the representation orbit from dependence on one selected parent, define

$$K_\rho(q) = T_0 + \rho\left(e^{2\pi i q}T_{+1} + e^{-2\pi i q}T_{-1}\right), \qquad 0 \le \rho \le 1. \tag{72}$$

Thus $\rho = 0$ is the onsite four-state parent and $\rho = 1$ is the full parent in Eq. (5). The onsite matrices, retained rank, frame protocol, and rotation family are unchanged. Every sampled ρ must independently clear the gap, transport-rank, reconstruction, and momentum-refinement controls.

10.2 Implemented metrics and interpretation of the parent-family figure

The implemented calculation concerns complete finite-mesh retained Hamiltonians, not range-truncated approximations. For each sampled parent scale ρ, the four-state parent is diagonalized, the lower two-dimensional spectral subspace is transported into a periodic orthonormal frame, and the resulting 2×2 retained field $H_\rho(q_i)$ is formed. The comparison field is

$$A_{\rho,a}(q_i) = U_a(q_i)^\dagger H_\rho(q_i) U_a(q_i), \qquad U_a(q) = R(a \sin 2\pi q). \tag{73}$$

Equation (73) is a pointwise unitary change of representation. The complete reference and comparison fields therefore have identical ordered eigenvalues in exact arithmetic. The reported band defect is

$$\Delta_{\mathrm{band}}(\rho, a) = \max_{i,n} \frac{|\varepsilon_n[A_{\rho,a}(q_i)] - \varepsilon_n[H_\rho(q_i)]|}{E_G}. \tag{74}$$

It is a numerical confirmation of the constructed similarity relation, not evidence that two independently derived material Hamiltonians agree. Exact zeros in Fig. 5(a) are placed at 10^{-18} only for logarithmic display.

Let $H^\circ_\rho(q_i)$ denote the traceless retained field and define

$$\kappa_\rho = \frac{1}{N}\sum_i \frac{\|H^\circ_\rho(q_i)\|_F^2}{2E_G^2}. \tag{75}$$

The continuous harmonic maximization gives the largest normalized correlation $\mu_{\rho,a,\max}$ attainable by one momentum-independent real rotation. The candidate also admits one positive splitting multiplier. Its irreducible matrix-loss threshold is

$$\epsilon_{\min}(\rho, a) = \sqrt{\kappa_\rho\,[1 - \max(0, \mu_{\rho,a,\max})^2]}. \tag{76}$$

Consequently, panel (b) plots $\epsilon_{\min}/\sqrt{\kappa_\rho}$ and panel (d) uses

$$x = \frac{\epsilon_{\min}^2}{\kappa_\rho}. \tag{77}$$

This metric answers a restricted representation question: how much of the momentum-dependent comparison field cannot be reproduced by one global real rotation, one positive scale, and the common mean-energy field. It is not a distance between abstract operators under arbitrary momentum-dependent unitary equivalence.

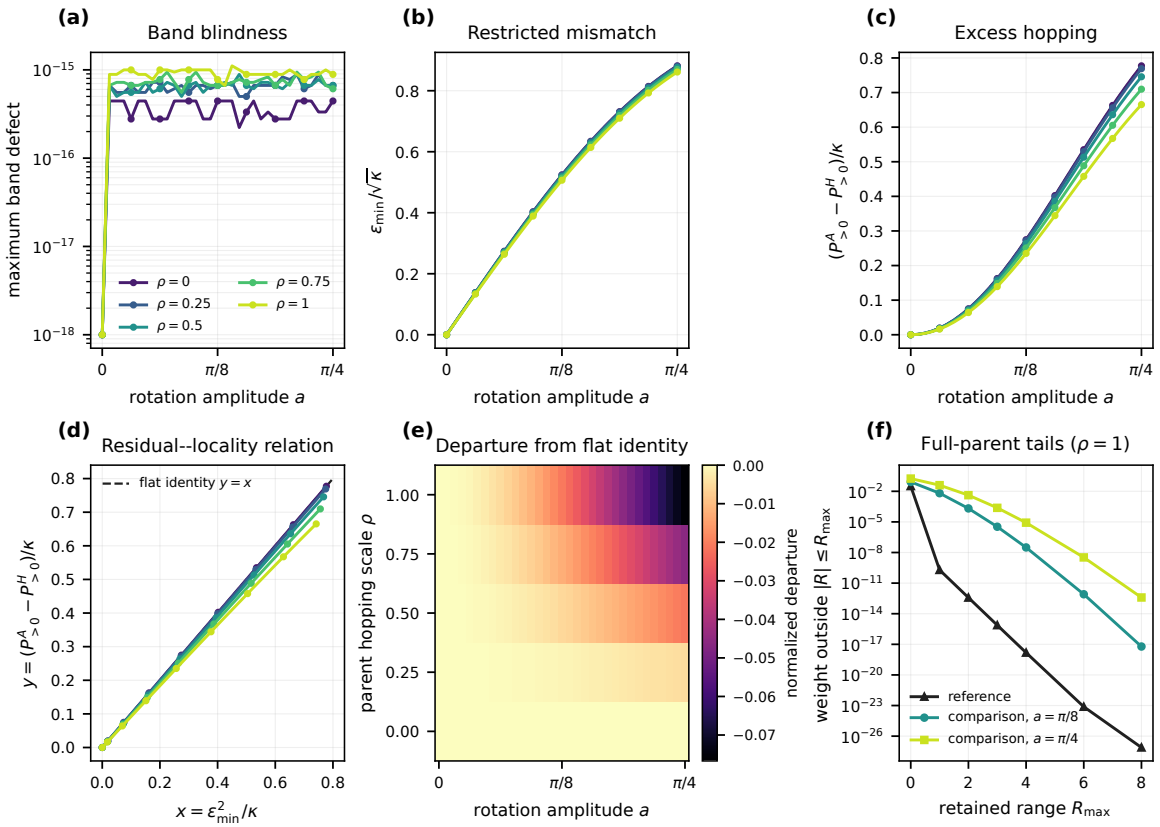


Figure 5: Band equivalence and locality across the sampled parent family. **(a)** Maximum ordered-eigenvalue defect; exact zeros are displayed at 10^{-18} on the logarithmic axis. **(b)** Minimum restricted-alignment threshold normalized by $\sqrt{\kappa}$; the dashed curve is the analytic flat-parent comparator. **(c)** Comparison-minus-reference non-onsite weight normalized by κ, with the flat analytic identity shown dashed. **(d)** Parametric relation between normalized squared operator residual x and normalized excess locality weight y; the dashed diagonal is the flat identity $y = x$. **(e)** Directly calculated departure $y - x$ over the 5×33 sampled parameter pairs. **(f)** Full-parent cumulative tails at selected amplitudes. Colors identify the parent hopping scales in panel (a), markers show every fourth amplitude sample, and connecting lines are guides to the eye. All five momentum studies through $N = 256$ are 'converged'; the finite ρ and amplitude inventories are not continuous-domain certificates. This is a pre-version development calculation, not a retained evidence run or material validation.

For centered finite-mesh cell displacements, the implementation constructs Fourier blocks $T_R^{H_\rho}$ and $T_R^{A_{\rho,a}}$. In the Wannier representation generated by the chosen periodic Bloch frame, their entries have the tight-binding meaning

$$[T_R^X]_{mn} = \langle 0m|\widehat{X}|Rn\rangle, \qquad X_{mn}(q) = \sum_R e^{2\pi i q R}[T_R^X]_{mn}, \tag{78}$$

where $|Rn\rangle$ is the Wannier orbital in cell R associated with retained orbital n. On the finite construction mesh this is a discrete, potentially aliased Wannier representation rather than a continuous localization certificate. A momentum-dependent Bloch-frame rotation corresponds to mixing Wannier orbitals across lattice cells. It can therefore redistribute the real-space matrix elements in Eq. (78) without changing the complete Bloch spectrum. The implementation assigns each block the per-retained-state power

$$p_R^X = \frac{\|T_R^X\|_F^2}{2E_G^2}, \qquad P_{>0}^X = \sum_{R\neq 0} p_R^X, \qquad X \in \{H_\rho, A_{\rho,a}\}. \tag{79}$$

Panel (c) plots the signed excess

$$y = \frac{P_{>0}^{A_{\rho,a}} - P_{>0}^{H_\rho}}{\kappa_\rho}. \tag{80}$$

Equivalently,

$$p_R^X = \frac{1}{2E_G^2}\sum_{m,n} |\langle 0m|\widehat{X}|Rn\rangle|^2. \tag{81}$$

Thus p_R^X is an entrywise squared-amplitude diagnostic for hopping between cell Wannier orbitals, summed over all retained-orbital pairs and normalized per retained state and by E_G^2. It is not a transition probability. Although both terms in the numerator are nonnegative, their difference is not a norm and need not be nonnegative for an arbitrary family. It is positive for the sampled family. At $\rho = 0$, the reference has no non-onsite hopping and the flat-parent theorem gives $y = x$. For $\rho > 0$, the reference already contains non-onsite blocks. Writing $T_R^A = T_R^H + \delta T_R$ shows that the excess contains both $\|\delta T_R\|_F^2$ and the cross term $2\,\mathrm{Re}\,\mathrm{tr}[(T_R^H)^\dagger \delta T_R]$. No theorem therefore identifies y with the restricted residual x for a dispersive parent. Panels (d) and (e) report the calculated relation and the diagnostic departure $y - x$ rather than assuming the flat identity. The systematic negative departure means that the increase in non-onsite block power accounts for less than the full normalized restricted residual in this family; it is not a failure of band equivalence.

Finally, panel (f) reports

$$P_{>R_c}^X = \sum_{|R|>R_c} p_R^X \tag{82}$$

for the full parent $\rho = 1$ at selected amplitudes. This is the hopping power that a symmetric range cutoff would omit. The implementation does not yet construct the truncated Hamiltonian or calculate its post-truncation bands, velocities, or effective masses. Thus panel (f) measures representation-dependent real-space compactness, not a spectral truncation error.

The figure matters for reduced solid-state modeling because the gauge of a retained subspace controls how economically its complete Bloch field can be represented by real-space hopping blocks. Panels (a)–(e) show that exact band equivalence can coexist with a large restricted representation residual and a redistribution of hopping power, and that this conclusion survives the sampled introduction of parent dispersion. They do not establish physical inequivalence of the complete gauge-related Hamiltonians. The parent homotopy is synthetic, the five ρ values and 33 amplitudes are finite direct samples, and the coefficients remain finite-mesh Fourier coefficients rather than a continuous-domain or material-localization certificate.

All five momentum studies return 'converged'. The smallest sampled external gap decreases from $1.3436583E_G$ at $\rho = 0$ to $1.0566461E_G$ at $\rho = 1$, while the reference non-onsite weight rises from numerical noise to 0.0323839. At $a = \pi/4$, the scale-normalized minimum threshold changes from 0.8815979 to 0.8613625, and the excess comparison non-onsite weight changes from 0.1431824 to 0.1437693 [Fig. 5(b)–(c)]. The normalized flat-identity departure $y - x$ changes from roundoff at $\rho = 0$ to -0.0767265 at $\rho = 1$ [Fig. 5(d)–(e)]. Thus the restricted mismatch and excess hopping remain large while the exact flat residual–tail identity deforms systematically with parent dispersion. The representation-induced mismatch is therefore observed throughout the sampled homotopy, not only for the endpoint parent. The five ρ values and 33 amplitudes remain finite samples; they do not establish a continuous-ρ bound or resolve amplitude interpolation.

11 Numerical-resolution study

11.1 Nested construction meshes

The proposed construction meshes are

$$N \in \{32, 64, 128, 256, 512\}, \qquad q_i = -\frac{1}{2} + \frac{i}{N}, \quad i = 0, \ldots, N-1. \tag{83}$$

Here N is the number of momentum samples in one reciprocal-space quadrature, not the number of independent model calculations. For example, $N = 64$ means that the single matrix-valued function $K(q)$ is assembled and diagonalized at 64 distinct momenta. The matrices differ because the phase $e^{2\pi i q R}$ differs at each momentum. No parent parameter is refitted between points.

The momentum grids are nested under doubling: every point on an N-point grid is an even-index point on the $2N$-point grid. Previously evaluated points may therefore be reused, and a complete refinement through $N = 512$ contains 512 unique construction momenta rather than the sum of all mesh sizes.

The completed flat-parent convergence study uses one fixed set of 65 amplitudes at every momentum resolution. Its 'converged' disposition concerns momentum refinement only. It compares finite-mesh maximum correlation, the two thresholds, non-onsite hopping weight, shared-block Fourier power, maximizing-angle status, the finite-mesh–Bessel quadrature defect, band defect, the three-way identity defect, and Parseval defect. The discrete $\mu_{N,\max}(a)$ and analytic $J_0(2a)$ remain separate. The amplitude interpolation study in Fig. 3(b) is an additional analysis and is not part of that momentum-convergence disposition.

The initial non-flat convergence implementation compares the sampled external gap, shared-node parent eigenvalues and retained spectral projectors, frame orthonormality and closure, Hermiticity, harmonic maximum and maximizing-angle status, $\mu_{N,\max}$, the two thresholds, Fourier-block powers, non-onsite weight, band defect, quadratic reconstruction, and Parseval consistency. The neighboring-frame transport singular value is a per-mesh rank-qualification lower bound, not a convergence observable. The study has no $J_0(2a)$ analytic comparator and may not assume the flat-parent three-way locality identity. Full parent eigenvalue and projector arrays, admissible-set quadratics, rectangle minima, witness coordinates, support points, containment margins, separation bounds, and post-decision evaluation losses remain prospective extensions. Its amplitude refinement must use a reviewed direct or adaptive rule that replaces the inadequate proposed ladder in Eq. (71).

For a scalar, vector, or matrix quantity z, the proposed continuous convergence test is

$$\|z_{\text{fine}} - z_{\text{coarse}}\|_\infty \le \epsilon_{\text{abs}} + \epsilon_{\text{rel}} \max(1, \|z_{\text{fine}}\|_\infty), \tag{84}$$

with draft values $\epsilon_{\mathrm{abs}} = 10^{-10}$ and $\epsilon_{\mathrm{rel}} = 10^{-8}$. These are deterministic numerical tolerances, not physical error bars or probabilistic confidence limits.

For the completed flat-parent momentum study, every implemented continuous channel must pass on two successive refinements and maximizing-angle status must remain unchanged on the corresponding three levels. For a final non-flat study, every continuous quantity must pass on two successive refinements, every discrete disposition must remain unchanged on three successive levels, and every decisive threshold or separation margin must be at least ten times larger than the corresponding observed refinement change. If the applicable momentum or amplitude cap is reached first, qualification fails. The largest attempted mesh is not called converged merely because no larger calculation was performed.

11.2 Disjoint post-decision evaluation

After the construction size N_* has been selected, the evaluation mesh has size

$$M = 4N_* + 1 \tag{85}$$

and coordinates

$$\widetilde{q}_\ell = -\frac{1}{2} + \frac{\ell + 1/(N_* + 1)}{M}, \qquad \ell = 0, \ldots, M-1. \tag{86}$$

The rational offset makes these coordinates distinct from the construction coordinates. A future retained implementation must additionally check all coordinate separations in the physical reciprocal unit against a fixed tolerance.

For the full non-flat workflow, the disposition, witness, alignment, and separation certificate are fixed before the evaluation mesh is examined. Reference and comparison matrices at evaluation coordinates are reconstructed from the finite Fourier representation obtained on the construction mesh. The selected candidate is then evaluated without refitting. This procedure probes interpolation and aliasing behavior between construction nodes. It does not create statistical independence, estimate experimental uncertainty, or validate the parent model. Neither current smooth implementation assigns a post-decision admissible-set disposition or executes this evaluation-mesh workflow.

12 Fourier reconstruction and locality

For the selected construction mesh, the matrix-valued forward transform is

$$T_R^{(N_*)} = \frac{1}{N_*} \sum_{i=0}^{N_*-1} e^{-2\pi i q_i R} H(q_i), \tag{87}$$

and the finite trigonometric interpolant is

$$H_{N_*}(q) = \sum_R e^{+2\pi i q R} T_R^{(N_*)}. \tag{88}$$

The identity $H_{N_*}(q_i) = H(q_i)$ holds at the construction nodes. Away from those nodes, H_{N_*} is a finite interpolant and is not substituted for the continuous field without an explicit interpolation or convergence check. The displacement inventory uses centered integer representatives of the finite cyclic mesh. For an even mesh, the Nyquist representative is $-N_*/2$; Hermitian symmetry there is interpreted cyclically, so its block must be Hermitian. The calculation and verifier must share this convention.

For maximum retained range R_c, the finite-mesh omitted-block locality norm is

$$\eta_{N_*}(R_c) = \left[\sum_{|R|>R_c} \left\| T_R^{(N_*)} \right\|_F^2 \right]^{1/2}. \tag{89}$$

It is evaluated separately for the reference matrix, comparison matrix, and selected aligned candidate. The reported value retains the parent energy unit. The calculation checks Fourier reconstruction, Parseval consistency under the declared normalization, and cyclic Hermitian hopping symmetry $T^{(N_*)}_{(-R) \bmod N_*} = (T^{(N_*)}_R)^\dagger$. Locality ranges are fixed before execution, listed in increasing order, and kept strictly below half the construction mesh.

These diagnostics describe the selected finite mesh and basis. They are not general claims of exponential localization, compressibility, or material accuracy.

13 Independent verification and reproducibility

13.1 Current numerical owners

The pre-version implementation consumes the exact revisions

`projectkoios-optimization`
`cf551409872a621b4e47d2236ac4ffdc12c187c0`
`projectkoios-physkit`
`07b2a1ee3ccd07926e90f43b4d1046efccc5ad8c`.

Optimization owns continuous harmonic maximization and each complex-unitary pairwise Procrustes solve in the producer. A numerically deficient overlap returns singular and rank evidence but no full unitary transformation; KSD then applies the stricter declared transport-overlap qualification gate. PhysKit owns immutable matrix-valued hopping records and raw block Frobenius powers. KSD supplies the finite-mesh identities, divides raw powers by the retained rank and E_G^2, checks Parseval consistency, and interprets the resulting locality channels.

PhysKit's hopping contracts use ordinary integer-lattice displacement semantics. The complete even-N Fourier inventory in this experiment remains cyclic and aliased: the centered Nyquist representative $-N/2$ is self-conjugate only modulo N. KSD therefore retains the cyclic Hermiticity check for the complete finite inventory. Any later ordinary-lattice truncation must use a conjugate-closed retained set with $R_c < N/2$, excluding the Nyquist representative. No owner API converts the finite coefficients into continuous hoppings without a convergence argument.

13.2 Independent numerical reconstruction

The implemented flat-parent verifier reads canonical input and result records but does not call the producing experiment action or trust a serialized disposition. It independently reconstructs the sampled matrices, band defect, second-harmonic coefficients, continuous maximum and unique-or-degenerate angle status, finite-mesh and analytic correlations, thresholds, Fourier-block powers, non-onsite weight, Parseval identity, and flat-parent three-way identity. Its convergence verifier separately reconstructs every adjacent momentum-mesh comparison and the final 'converged' or 'unresolved' disposition.

The implemented 'SmoothFourStateVerifier' independently reconstructs Eq. (5), repeats the eigensystem and frame checks, reconstructs the comparison field, Fourier blocks, Parseval quantities, and original restricted-alignment loss, and obtains the harmonic maximum directly as $b_0 + \sqrt{c^2 + d^2}$ without calling the producer's Optimization maximizer. A future full verifier must additionally rebuild Q, g, and c from Eq. (49); recompute rectangle minima; check witness membership; derive support points through linear solves; verify containment; reconstruct lower and upper separation bounds; and apply Eqs. (87)–(89) to disjoint evaluation channels. Those prospective certificate checks are not claimed as current four-state coverage.

Verification reports contain channel-resolved maximum defects rather than only a Boolean pass flag. Dimensionless defects and energy-valued defects remain separate. The main calculation and verifier may share binary64 arithmetic and the same numerical library; that dependence is disclosed. Independence here means separate scientific reconstruction and control flow, not independent hardware or an unrelated eigensolver.

13.3 Serialization and evidence

The implemented flat-parent input, result, and verification records use strict JSON schemas that reject unknown fields. Non-flat schemas are intentionally deferred until its numerical contract is reviewed. Numeric strings and Boolean values presented as real numbers are rejected by the applicable public contracts. Floating-point values must be finite. Complex values are represented by explicit real and imaginary components, and arrays carry fixed dimensions wherever shape is part of the scientific contract.

The active result records the SHA-256 digest of the exact canonical input bytes. The verification record binds both input and result digests. Family, schema, and method identities are checked against expected values rather than trusted from the payload, and the input scope explicitly states that the calculation is not retained. Any change to a nested value or immutable identity either changes the digest or fails schema validation. A future retained schema must additionally bind immutable version and run identities; no such identity is inferred for the current pre-version records.

A retained run directory will be created by an exclusive no-overwrite operation. It will contain canonical input, result, and verification records; the complete convergence table; source and dependency provenance; platform and command information; authenticated schema identities; the exact source manifest; a checksum manifest; and a readable report separating controls, calculated quantities, and interpretation. Experiment version, run identity, and content-addressed evidence-package identity remain distinct.

13.4 Failure policy

The active flat-parent calculation fails closed on nonfinite data, invalid dimensions, malformed amplitudes or reciprocal meshes, nonpositive convergence thresholds, failed harmonic reconstruction, failed Fourier or Parseval identities, schema violations, digest mismatch, or disagreement with the verifier. The initial non-flat calculation additionally fails closed on inadequate external gap, rank-deficient transport, failed frame closure, inconsistent loss reconstruction, failed Fourier nodal reconstruction, Parseval inconsistency, or disagreement with its verifier. The future expanded contract must also fail closed on nonpositive raw weights, malformed or reused evaluation coordinates, indefinite quadratics, invalid certificates, and checksum mismatch. Such failures do not become scientific outcomes.

An empty admissible component, an invalid containment premise, an insufficient separation margin, or failure to find a common witness can be a legitimate scientific non-result. These conditions lead to an empty-component record, an invalid-premise record, or an unresolved disposition rather than a software success being reinterpreted as evidence.

14 Prespecified controls and execution status

Table 1 separates the mathematical design already stated in this manuscript from controls that still require final scientific review. Confirmatory execution is not permitted until every open control has a value, unit, rationale, and canonical representation.

Table 1: Status of outcome-relevant controls. "Specified" means that the value or rule appears in this manuscript. "To be fixed" means that no confirmatory calculation may proceed until the control is approved and recorded.

Control	Status	Location or required record
Parent hopping model and reciprocal convention	Specified	Eqs. (5) and preceding matrices
Retained rank and frame method	Specified	Eqs. (10)–(13)
Minimum external gap	To be fixed	Positive energy-valued lower bound and rationale
Minimum transport-overlap singular value	Draft value only	Final σ_{rank} and tolerance
Comparison rotation field	Specified	Smooth family in Eq. (17)
Candidate and continuous alignment families	Specified	Eqs. (19)–(38)
Momentum construction and evaluation rules	Specified	Eqs. (83)–(86)
Amplitude refinement rule	To be fixed	Direct analytic or reviewed adaptive rule
Parameter rectangle	To be fixed	Four dimensionless bounds
Spectral and operator thresholds	To be fixed	Values, units, and scientific status
Separation resolution	To be fixed	ρ_{sep} and its rationale
Locality ranges	To be fixed	Ordered integer ranges
Numerical tolerances	Draft values only	Channel-specific finalized table
Software revisions	Development known; retained open	Dependency lockfiles; exact retained manifests
Confirmatory execution authorization	Not granted	Separate operator decision

15 Interpretation and limitations

Theorem 1 is not specific to the numerical parent matrices. It applies to any sampled rank-two Hermitian field with positive weights, a comparison target related by unitary conjugation, and the trace–traceless candidate family in Eq. (19). The parent, momentum dependence, and continuous global alignment enter the optimized geometry through κ and $\mu_{\max}$.

An instance-specific conclusion applies only to the stated parent model, retained rank-two subspace, frame convention, comparison representation, candidate family, parameter rectangle, continuous global real-rotation family, thresholds, numerical resolution, and data roles. A compatible result would show that one parameter pair meets the two declared limits in this controlled setting. A separated result would show that the two admissible regions are farther apart than the declared resolution under the verified geometric premises. Neither result would establish material validity, parent-model correctness, uncertainty quantification, uniqueness, continuum convergence, or adequacy under unrestricted momentum-dependent unitaries.

The frame, matrix loss, and locality norms are basis dependent by design. The spectral loss is basis invariant. If the full generating $U_a(q)$ is accepted as the state-space identification, $H(q)$ and $A_a(q)$ are unitarily equivalent representations and their abstract-operator difference is zero. The reported nonzero residual instead answers the declared restricted question in which only one momentum-independent real rotation may identify retained coordinates. Keeping these channels and identification rules separate is essential to the scientific question rather than a technical inconvenience. Likewise, convergence of the spectrum does not imply convergence of represented matrices, hopping blocks, or geometric certificates; each applicable channel is therefore tested explicitly.

No Quantum ESPRESSO or Wannier90 calculation is part of this controlled protocol. A material application would require a separate authorized design specifying structures, pseudopotentials, exchange–correlation approximation, cutoffs, meshes, Wannier windows and projections, energy alignment, resource estimates, and provenance.

References

[1] Peter H. Schönemann. A generalized solution of the orthogonal procrustes problem. *Psychometrika*, 31(1):1–10, 1966.

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[3] NIST Digital Library of Mathematical Functions. Bessel and hankel functions: Generating function and associated series. Section 10.12, Jacobi–Anger expansions.

A Exact rectangle minimization

For

$$f(p) = (p - p_c)^T Q (p - p_c) + m, \qquad Q = \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{12} & Q_{22} \end{pmatrix},$$

the unconstrained center is included when it lies in X. On a vertical edge $p_1 = b$, the unconstrained edge minimizer is

$$p_2 = p_{c,2} - \frac{Q_{12}}{Q_{22}}(b - p_{c,1}),$$

which is clipped to $[\lambda_{\min}, \lambda_{\max}]$. This is done for $b = s_{\min}$ and $b = s_{\max}$. On a horizontal edge $p_2 = b$, the corresponding minimizer is

$$p_1 = p_{c,1} - \frac{Q_{12}}{Q_{11}}(b - p_{c,2}),$$

clipped to $[s_{\min}, s_{\max}]$. Direct evaluation of the original quadratic at these points yields the exact rectangle minimum for a positive-definite two-dimensional quadratic. Corners appear automatically when clipping reaches an endpoint.

B Support and containment checks

For a general quadratic ellipse

$$\mathcal{E} = \{p : (p - p_c)^T Q (p - p_c) \le \beta\}, \qquad Q \succ 0,$$

the support point and support value in direction $d \neq 0$ are

$$p_+(d) = p_c + \sqrt{\frac{\beta}{d^T Q^{-1} d}} Q^{-1} d, \tag{90}$$

$$\max_{p \in \mathcal{E}} d^T p = d^T p_c + \sqrt{\beta\, d^T Q^{-1} d}. \tag{91}$$

Coordinate extrema are obtained using $d = (1, 0)^T$ and $d = (0, 1)^T$ together with their negatives. Containment requires all four extrema to remain within the corresponding box bounds after accounting for the fixed containment tolerance. The support calculation uses linear solves with Q and never substitutes a box-clipped point into a formula for the full ellipse. For the common curvature in Theorem 1, these checks reduce to the semiaxes stated by Eqs. (39) and (40).

A candidate upper bound on the distance is computed from an explicitly feasible pair (p_s, p_o) as $\|p_s - p_o\|_2$. A projected lower bound is accepted only after its support points, component orientation, containment premise, and threshold membership have been independently reconstructed. The inequalities

$$0 \le \underline{\delta} \le \delta^* \le \overline{\delta}$$

must hold within numerical tolerance. A violation indicates a calculation or premise failure, not an unusual scientific result.

C Execution sequence

The confirmatory calculation follows the sequence below without adaptive changes to the scientific controls.

1. Parse canonical input and reject unknown, malformed, nonfinite, or inconsistent fields.
2. Construct $K(q)$ on every mesh in the declared refinement ladder.
3. Diagonalize the parent, retain the lowest two states, and verify the external gap and retained projectors.
4. Build and periodically close the transported frame; stop on deficient overlap.
5. Construct $H(q)$ and the complete amplitude family $A_a(q)$ and retain all parent diagnostics.

6. For every momentum and amplitude mesh, construct the spectral quadratic and the exact second-harmonic operator correlation from the original residuals.

7. Maximize the continuous harmonic over its half-open fundamental interval, record unique or degenerate angle status, construct the optimized operator quadratic, and minimize both quadratics over the fixed parameter rectangle.

8. Apply all continuous and discrete convergence gates. Stop with failed qualification if the declared caps are reached first.

9. Freeze the selected construction resolution without changing thresholds or parameter bounds.

10. Search for a common witness and verify it from the original residuals.

11. If no witness is established, test every separation-certificate premise and construct conservative lower and upper bounds.

12. Assign the disposition from construction data and the declared numerical uncertainty rules.

13. Generate the disjoint evaluation mesh and compute evaluation losses without changing the disposition.

14. Transform the reference, comparison, and selected candidate matrices to hopping blocks and calculate range-resolved locality norms.

15. Serialize canonical evidence, perform independent reconstruction, validate schemas and digests, and retain the package only if all required checks pass.